

Auditable Skill-Layer Robotics for Factory Sorting: Official-Definition Alignment and a Recomputed 100/100 Objective Score

Technical Report

Team BIPT-EDU

Submission revision: August 3, 2026

Official reference commit: [129e94a9](#)

Recomputed objective score: **100/100**

Abstract

This report describes a unified RobotAgent for the five JCIOT 2026 RunningRobot factory-sorting levels and, equally importantly, an auditable chain from the official task definition to the submitted trajectory bytes. The previous package can be reproduced as exactly 65/100 under the task definition at commit [129e94a9](#): L3 used the obsolete `input_6` / left-blue route and L5 released at obsolete `output_6`. The corrected system binds execution to the current official routes, synthesizes grasp approach frames online from official object sites, performs clearance-aware base positioning entirely in the editable skill layer, and maintains L5 state continuity across three physical grasp/place cycles. Five packaged trajectories recompute to 10, 15, 20, 25, and 30 points, with no recorded collision flags. We deliberately distinguish this deterministic local rescore from the organizer’s final code-audited result. No trajectory JSON, locked task configuration, transport attachment, or official environment file is post-processed by participant code.

Contents

1	Evaluation Target and Root Cause	2
1.1	Pinned official task definition	2
1.2	Why the previous package scored 65	2
2	System Architecture and Edit Boundary	2
2.1	Compliance boundary	3
3	Method	4
3.1	Exact auxiliary-station resolution	4
3.2	Hybrid grasp with runtime geometry synthesis	4
3.3	Clearance-aware staging in the skill layer	5
3.4	L5 transaction-scoped state continuity and placement	6
4	Novelty Statement	7
5	Experimental Protocol	8
5.1	Trajectory selection and scorer	8
5.2	Integrity controls	8
6	Results and Analysis	8
6.1	Objective scores	8
6.2	Recorded physical evidence	9

7	Reproducibility and Traceability	9
8	Limitations	10
9	Conclusion	10
A	Submission Package Contents	11

1 Evaluation Target and Root Cause

1.1 Pinned official task definition

All task selection in this revision is pinned to official commit [129e94a9](#). The relevant source is the official `JCIIOT/knowledge/task_config.json`; Case 3’s Placement Point 1 wording is also consistent with the official erratum. Table 1 records the exact routes and accepted objects used by both execution and verification.

Level	Source	Target	Accepted object candidates	Max
L1	input_5	output_4	line_5_container_h01_near/far	10
L2	input_6	output_4	green_tote_b01_upper/lower	15
L3	aux_input_1	output_5	blue_tote_b01_far_right/near_right	20
L4	input_2	output_5	blue_container_h01_back_upper/lower	25
L5	input_1	aux_output_1	three white_tote_b01_left_* objects	30

Table 1: Official-current objective task definition used in this report.

1.2 Why the previous package scored 65

The mismatch is deterministic rather than stochastic. The old self-test used stale routing and an unsafe nearest-object fallback; the current official definition rejects those substitutions. Table 2 separates configuration failure from manipulation failure.

Level	Previous	Final	Evidence-based explanation
L1	10	10	Route and accepted object already matched the current definition.
L2	15	15	Route and accepted object already matched the current definition.
L3	0	20	Previous trajectory grasped <code>blue_tote_b01_near_left</code> at <code>input_6</code> ; current source is <code>aux_input_1</code> and candidates are the two right-side blue totes.
L4	25	25	Route and accepted object already matched the current definition.
L5	15	30	All three grasps were valid, but previous releases were near <code>output_6</code> ; current target is <code>aux_output_1</code> .
Total	65	100	L3 recovered 20 points and L5 recovered 15 points.

Table 2: Exact reproduction of the 65-point result and the corrected objective rescore.

2 System Architecture and Edit Boundary

The system keeps semantic planning and physical execution separate. Qwen2.5:7B produces a short move/grasp/move/place plan, while deterministic skills resolve official station names, choose objects, plan base motion, and call the official physics backend. The verifier is outside the runtime loop and reads only the final ZIP bytes.

Unified RobotAgent Architecture

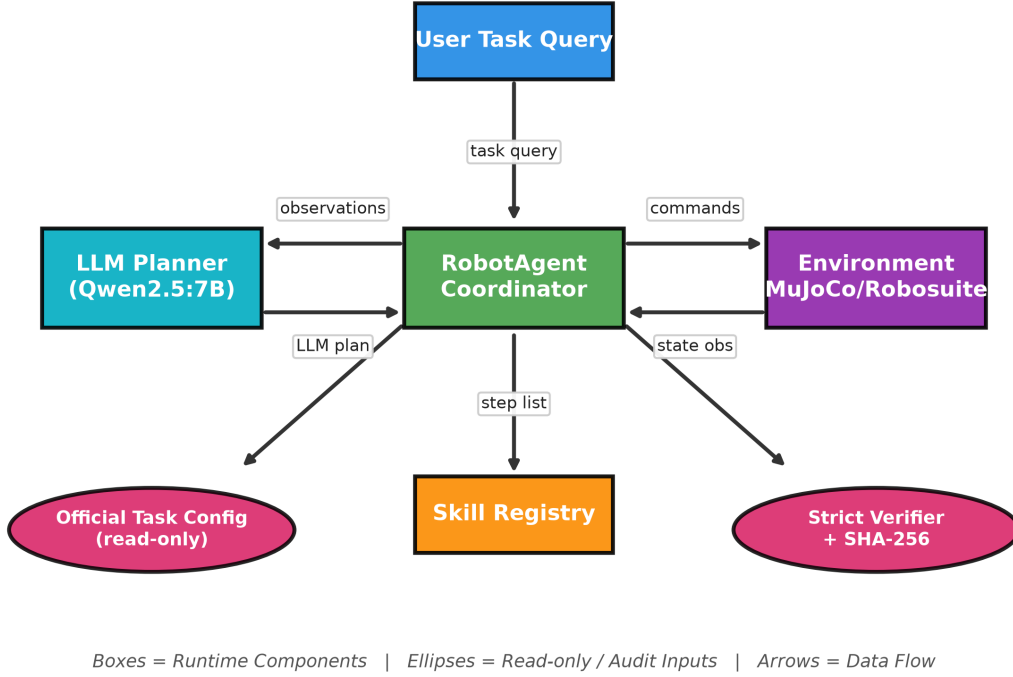


Figure 1: RobotAgent architecture with the official task configuration treated as read-only input and verification isolated from execution.

2.1 Compliance boundary

The official rules identify `src/robot_agent/core/`, `src/robot_agent/environments/`, `app.py`, and `knowledge/task_config.json` as forbidden edit areas. A direct Git content comparison against commit [129e94a9](#) shows no change in those paths. The same check finds no content change in the shared robosuite files or generated maps present in this submission. This wording deliberately ignores checkout-only LF/CRLF conversion on Windows. The navigation extension that an earlier draft placed in `core/map_loader.py` and `core/navigation.py` has been relocated to `skills/move.py`; the core files are restored to their official versions.

Area	Final status	Role
Official core, environments, app, task config, maps	No Git content diff from pinned official commit	Evaluation/runtime baseline
<code>src/robot_agent/skills/</code>	Participant edits	Station resolution, geometry, staged navigation, multi-object continuity
<code>knowledge/robot_params.json</code>	Participant-editable parameters	Grasp, lift, turn, and recorder settings
<code>pipeline/</code> and root verifier	Added tooling	Canonical runs, strict rescoring, packaging, and integrity reports

Table 3: Final implementation boundary.

3 Method

3.1 Exact auxiliary-station resolution

Auxiliary names contain ordinary station names as substrings: for example, `output_1` is contained in `aux_output_1`. The previous substring-first resolver could therefore navigate to the wrong station even when the LLM emitted the correct target. The final resolver first tests an exact known name, then tests embedded names in decreasing length order. This small ordering invariant is essential for both L3 and L5 and requires no task-config change.

Listing 1: Exact-first station resolution in the editable skill layer.

```
1 names = scene.all_port_names()
2 exact = target.strip()
3 if exact in names:
4     return station_goal(exact)
5 for name in sorted(names, key=len, reverse=True):
6     if name in target:
7         return station_goal(name)
```

3.2 Hybrid grasp with runtime geometry synthesis

The official BC checkpoint remains the first attempt in fallback mode. When it fails outside its familiar geometry, a deterministic expert uses the official collector’s lift, XY approach, descent, settle, close, and lift primitives. Event recording and official transport attachment remain in the unchanged backend.

The base approach is computed from live model geometry rather than written into the locked task configuration. Let object centre be c , and let s_R, s_L be the official right and left grasp sites. The measured outward direction and base pose are

$$u = \frac{(s_R + s_L)/2 - c}{\|(s_R + s_L)/2 - c\|}, \quad b = c + du, \quad d = 0.941 \text{ m}. \quad (1)$$

For objects whose natural standing side is blocked, the skill rotates this frame at runtime:

$$u' = R_z(\theta)u, \quad p_{R,L} = c + fu' \pm \ell R_z(90^\circ)u', \quad (2)$$

where $f = 0.315$ m, $\ell = 0.11$ m, $\theta = +90^\circ$ for the selected L2/L5 totes, and $\theta = -90^\circ$ for the selected L3 right-side tote. The points p_R, p_L are controller targets for the current grasp call; they do not rewrite XML sites, semantic maps, task configuration, or trajectory data.

Runtime Geometry-Derived Grasp Approach

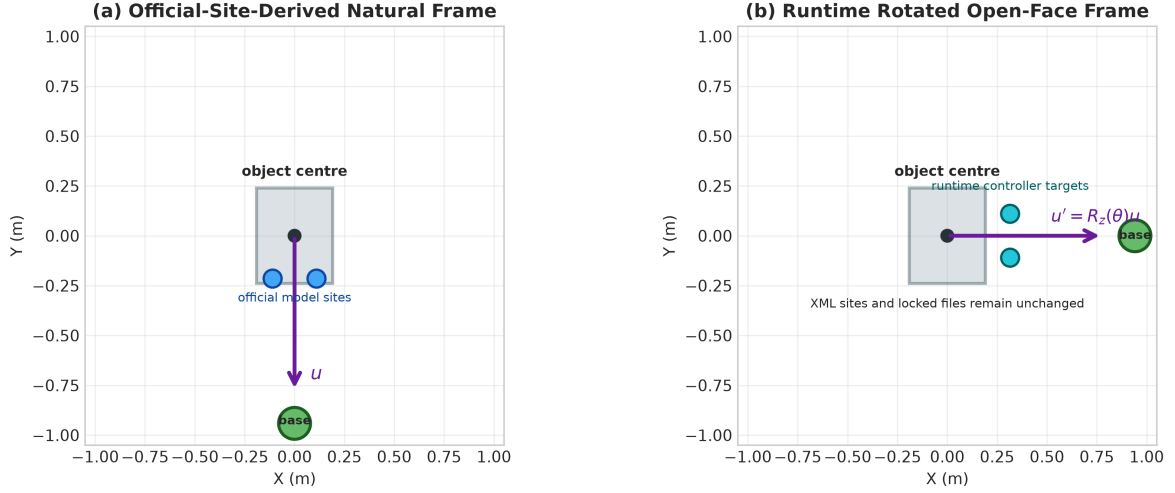


Figure 2: Runtime site-relative frame synthesis. The right panel illustrates a 90° rotation; the same formula supports either sign.

3.3 Clearance-aware staging in the skill layer

Navigation uses the official occupancy grid and official A* primitives. The editable MoveSkill creates temporary in-memory dilations with margins 0.45, 0.30, 0.15, and 0 m. Start and goal neighborhoods are restored from the official grid so valid table approaches are not removed. The first feasible margin is selected; the final stage is exactly the unmodified official grid.

For grasp positioning, the robot first drives to a staging point 0.65 m behind the computed base pose, turns with the official turn controller, and then follows a short straight physical approach. L5 keeps an additional 0.10 m table clearance. This prevents an extended arm from sweeping through a table during a large turn and preserves the base/object relationship captured by the official backend.

Skill-Layer A* Clearance Ladder with Endpoint Exemption

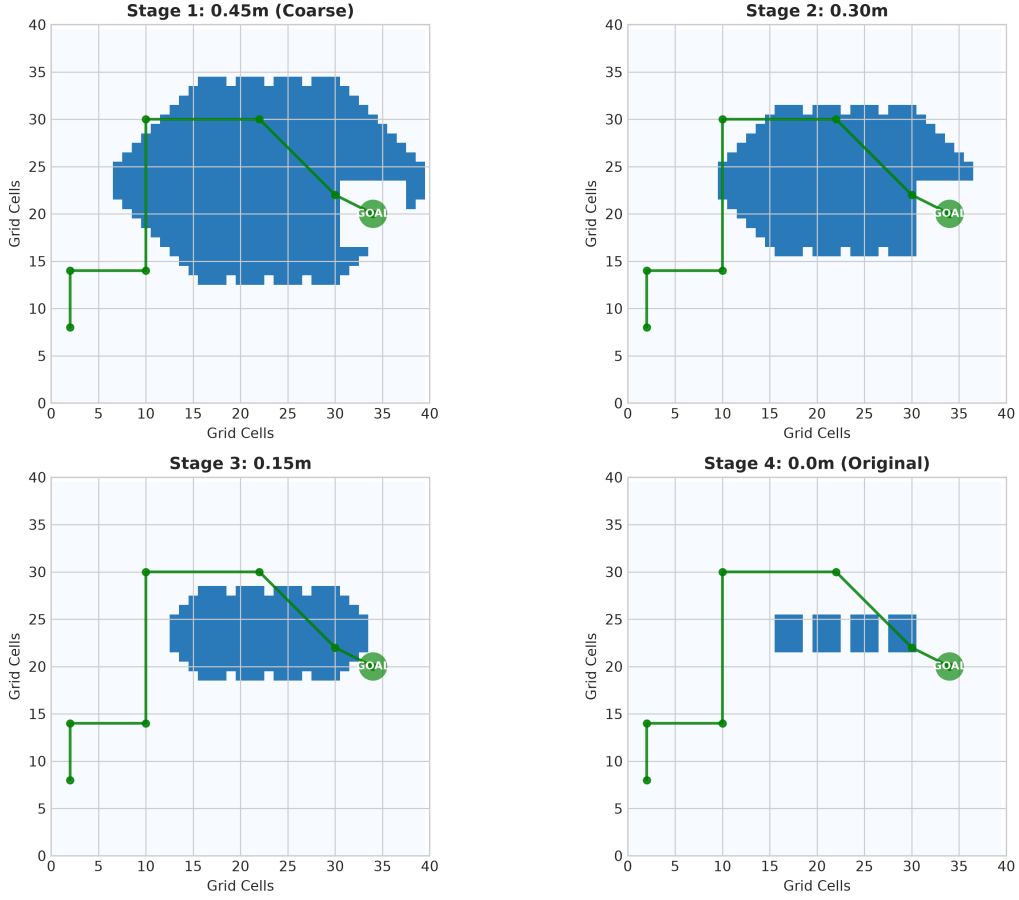


Figure 3: Illustrative skill-layer clearance ladder. Dilations are temporary planning views; the official occupancy array is never overwritten.

3.4 L5 transaction-scoped state continuity and placement

The official grasp helper creates a wrapped environment and synchronizes all material objects back to the navigation environment after a grasp. During a three-object task, that blanket synchronization can overwrite the physical result of earlier placements. The `PickUpSkill` therefore snapshots only non-target free-joint states before the official grasp call and restores those same non-target states afterward. It does not move the selected target, declare a grasp successful, or write transport attachment state; success and attachment remain controlled by the official grasp/lift pipeline.

Placement is similarly scoped. The three calls use the target centre and lateral offsets of 0 m, +0.38 m, and -0.38 m. Only the temporary station centre consumed by the official place controller is changed, and it is restored immediately after each call. Final positions are determined by physics and recorded normally.

L5 Target Alignment from Recorded Trajectories

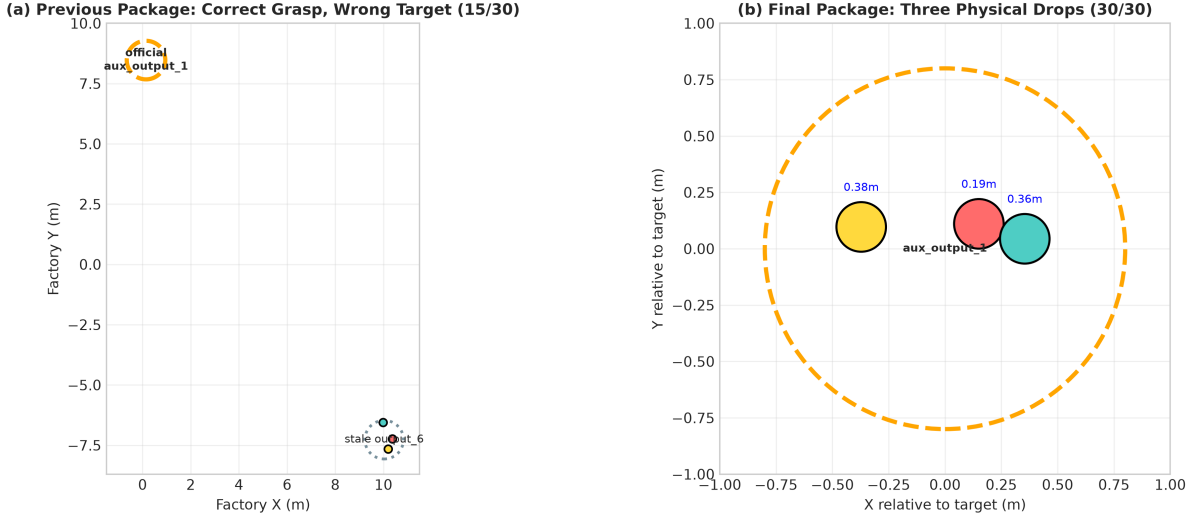


Figure 4: Recorded L5 evidence. Left: the old trajectory’s three objects were physically near stale output_6. Right: final object positions relative to official aux_output_1; distances are measured from the final trajectory frame.

4 Novelty Statement

The innovation claim in this revision is intentionally narrower and more auditable than the earlier draft. We do not claim a new imitation-learning algorithm, a general proof of safety, or state-of-the-art performance beyond this benchmark. The contribution is a set of composable skill-layer mechanisms that bridge an official learned/physics baseline to multi-scene execution while preserving the evaluation boundary.

1. **Runtime site-relative grasp-frame synthesis.** Official model sites become measurements rather than files to patch. One geometric formulation produces both natural and rotated controller targets and is exercised in L2, L3, and L5.
2. **Physically consistent two-stage base positioning.** A temporary clearance ladder, a safe staging pose, an official physical turn, and a straight final approach are combined inside allowed skills. The approach addresses both collision exposure and attachment-frame consistency without writing attachment data.
3. **Transaction-scoped multi-object continuity.** Non-target state preservation around the official wrapped grasp call prevents later grasps from erasing earlier placements. Three targettable points then convert a repeated single-object primitive into a stable three-object sequence.
4. **Versioned evidence chain.** One check binds the task commit, accepted candidates, ZIP members, raw/package hashes, and recomputed scores. This exposes stale-task success and trajectory post-processing.

Baseline gap	Added mechanism	Direct evidence
BC distribution shift	Site-relative deterministic fallback	Successful official grasp/lift events for selected L2–L5 objects
Table-adjacent approach	Skill-local clearance ladder and staged physical approach	Zero collision markers in all five submitted trajectories
Wrapped-env reset during L5	Non-target state transaction and three release points	Three named grasp events and three target placements
Stale local scoring	Commit-pinned strict verifier and SHA-256 report	Exact reproduction of 65, followed by 100 on the final set

Table 4: Scope and evidence for the revised innovation claims.

5 Experimental Protocol

5.1 Trajectory selection and scorer

The canonical set contains one successful simulator trajectory per level. L1, L2, and L4 retain valid earlier trajectories because their routes did not change. L3 and L5 were rerun against the corrected official routes. The strict verifier opens each ZIP, requires exactly `trajectory.json`, `score.json`, and `submission_manifest.json`, checks the manifest commit, hashes both package and trajectory bytes, and recomputes the score using the pinned task candidates.

The verifier does not run an LLM, MuJoCo, or rendering. It evaluates the recorded state sequence. This is useful for deterministic auditing, but it is not a substitute for the organizer’s final source-code review.

5.2 Integrity controls

- Packaged `trajectory.json` is a byte-for-byte copy of the raw simulator `trajectory*_OK.json`.
- The packaging script writes only `score.json` and `submission_manifest.json`; it does not rewrite trajectory frames.
- No participant skill writes the official transport-attachment structure.
- Full machine-readable report: `verification_report.json` in the final-package directory.
- The all-in-one archive SHA-256 is `a866718e44314ce5ffe7844a3b8a4d8b060765c077a8b6ff056d0d19a33cb8c0`.

6 Results and Analysis

6.1 Objective scores

Level	Maximum	Previous	Final	Delta
L1	10	10	10	0
L2	15	15	15	0
L3	20	0	20	+20
L4	25	25	25	0
L5	30	15	30	+15
Total	100	65	100	+35

Table 5: Deterministic objective rescore under the pinned official task definition.

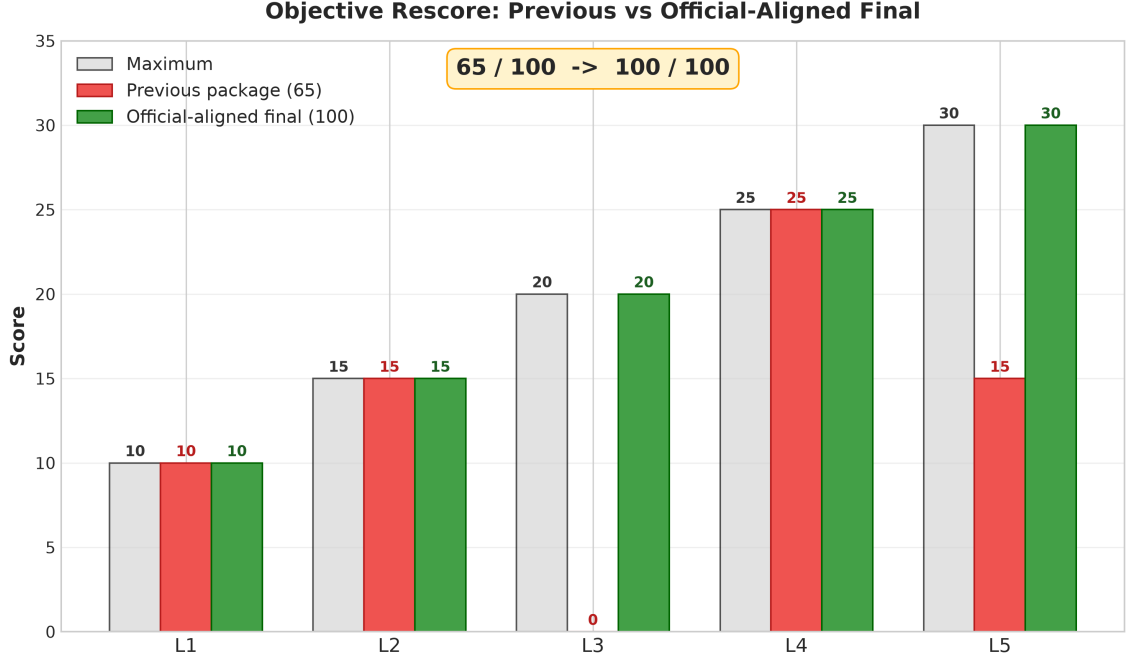


Figure 5: Previous and final objective scores. The changed levels match the two stale-route defects identified in Table 2.

6.2 Recorded physical evidence

Table 6 reports values directly extracted from the final trajectory JSON and score documents. Duration is final frame time minus first frame time; it is not wall-clock benchmark time. No submitted JSON contains `judge_collision_detected` or a true collision marker.

Level	Recorded selected object(s)	Frames	Span (s)	Final target evidence
L1	line_5_container_h01_near	1308	66.25	0.11 m from target
L2	green_tote_b01_lower	1153	52.05	0.09 m from target
L3	blue_tote_b01_far_right	1915	110.87	0.05 m from target
L4	blue_container_h01_back_upper	1842	85.20	0.15 m from target
L5	centre, front, back left white totes	8211	438.03	0.19, 0.36, 0.38 m from target

Table 6: Measurements extracted from the packaged trajectories.

The old-to-final score change is not presented as a controlled ablation of every algorithmic component. It is an end-to-end correction with two known causal defects: task-definition alignment for L3 and target alignment plus multi-object execution for L5. This distinction avoids attributing unsupported numeric gains to individual heuristics.

7 Reproducibility and Traceability

From the repository root, the final set is checked with:

Listing 2: Deterministic verification command.

```
1 JCI10T\.venv\Scripts\python.exe verify_submission.py \
2 --submissions submission_100_final_129e94a9
```

Expected output is five **PASS** rows and **TOTAL 100/100**. Key implementation-to-claim links are:

- `skills/scripted_grasp.py`: live and rotated approach geometry, deterministic expert fall-back;

- `skills/move.py`: exact auxiliary-station matching and temporary clearance ladder;
- `skills/pick_up.py`: staging, official turn, straight approach, non-target state transaction;
- `skills/place_down.py`: temporary three-point target selection;
- `pipeline/score_trajectory.py`: strict official-current candidate scoring;
- `verify_submission.py`: member, manifest, score, and SHA-256 checks.

8 Limitations

1. The reported 100/100 is a deterministic rescore of one selected trajectory per level, not a repeated-trial success-rate estimate.
2. The rotated approach constants are validated on the named competition objects and are not claimed to generalize to arbitrary geometry without additional collision perception.
3. Non-target state restoration compensates for the official wrapper’s blanket synchronization. A future backend API that synchronizes only the grasped target would remove this skill-level transaction.
4. The task is evaluated in static maps. Dynamic obstacles would require online local replanning.
5. Final acceptance remains subject to organizer code audit; this report documents evidence and boundaries rather than asserting organizer approval in advance.

9 Conclusion

The central lesson of this submission is that the 65-point discrepancy was not evidence of trajectory editing; it was the exact consequence of stale official-task routing in L3 and L5. The corrected system reaches a recomputed 100/100 while keeping official core, environment, task, and map files unchanged. Its technical contribution is not an inflated SOTA claim, but an auditable set of skill-layer mechanisms: runtime grasp-frame synthesis, clearance-aware staged base motion, transaction-scoped multi-object continuity, and integrity-bound verification. Each claim in this revision maps to current code or recorded trajectory evidence.

References

- [1] JCIOT 2026, “RunningRobot official repository,” commit [129e94a9](#). <https://github.com/JCIOT2026/JCIOT2026/tree/129e94a9cff787031472045e19c24a4baeaefc48>
- [2] T. Osa, J. Pajarinen, G. Neumann, et al., “An Algorithmic Perspective on Imitation Learning,” *Foundations and Trends in Robotics*, vol. 7, no. 1–2, 2018.
- [3] P. Hart, N. Nilsson, and B. Raphael, “A Formal Basis for the Heuristic Determination of Minimum Cost Paths,” *IEEE Transactions on Systems Science and Cybernetics*, 1968.
- [4] J. Schulman, J. Ho, A. Lee, et al., “CHOMP: Gradient Optimization Techniques for Efficient Motion Planning,” *IEEE ICRA*, 2009.

A Submission Package Contents

Each level ZIP contains exactly:

1. `trajectory.json` - raw simulator state log copied byte-for-byte;
2. `score.json` - objective breakdown under the pinned strict scorer;
3. `submission_manifest.json` - level, environment, method, and official commit metadata.

The five package hashes are listed in `submission_100_final_129e94a9/SHA256SUMS.txt`; raw trajectory hashes and per-level scores are listed in `verification_report.json`.